

Fake News Detection Using LSTM Networks and Pre-trained BERT Models

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GitHub: <https://github.com/TarifEKhan/Fake-news-detection>

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1. Introduction

Background

The rapid proliferation of online news and social media platforms has made the spread of misinformation easier than ever before. Fake news — fabricated or deliberately misleading content presented as factual journalism — has become a critical societal problem with measurable consequences for public health, political discourse, and social stability. High-profile cases of misinformation surrounding elections, public health crises, and financial markets underscore the urgency of developing automated detection systems.

Natural Language Processing (NLP) and deep learning offer promising avenues for addressing this challenge. By training models on large corpora of labeled news articles, we can build classifiers capable of distinguishing real from fake content with high accuracy, providing a scalable complement to human fact-checkers.

Objectives

- Build and evaluate a custom Bidirectional LSTM (BiLSTM) classifier for fake news detection.
- Fine-tune a pre-trained BERT transformer model on the same task.
- Establish baseline performance with traditional ML models (SVM, Random Forest).
- Compare all models across Accuracy, Precision, Recall, and F1-Score.
- Analyze the accuracy-efficiency tradeoff between simple and complex models.

Significance

Automated fake news detection supports fact-checking pipelines, content moderation on social platforms, and media literacy tools. This project contributes to that ecosystem by demonstrating that both lightweight (LSTM) and heavyweight (BERT) models can achieve high performance, with the choice depending on available computational resources.

2. Literature Review

Traditional Machine Learning Approaches

Early fake news detection systems relied on hand-crafted features combined with classical classifiers. Naive Bayes and Support Vector Machines (SVMs) with TF-IDF representations achieved reasonable accuracy on benchmark datasets. Castillo et al. (2011) demonstrated that credibility signals embedded in text and metadata could be exploited for classification. However, these methods struggle to capture long-range semantic dependencies in text.

Deep Learning Approaches

Recurrent Neural Networks (RNNs) and their gated variants — LSTMs and GRUs — significantly improved performance by modeling sequential context. Wang (2017) introduced the LIAR benchmark dataset and showed that LSTM-based models outperform SVMs on multi-class fake news classification. Bidirectional LSTMs further improved results by capturing both forward and backward context simultaneously.

Convolutional Neural Networks (CNNs) have also been applied to text classification, capturing local n-gram features. Hybrid CNN-LSTM architectures that combine local feature extraction with sequence modeling have shown competitive performance on several NLP benchmarks.

Transformer-Based Models

The introduction of BERT (Bidirectional Encoder Representations from Transformers) by Devlin et al. (2019) marked a paradigm shift in NLP. Pre-trained on massive corpora using masked language modeling and next-sentence prediction, BERT achieves state-of-the-art performance when fine-tuned on downstream tasks with minimal task-specific architecture changes. Kula et al. (2020) demonstrated that fine-tuned BERT models consistently outperform LSTM-based approaches on fake news datasets, though at significantly higher computational cost.

Key Insight: The Accuracy-Efficiency Tradeoff

A recurring theme in the literature is the tradeoff between model complexity and performance. While BERT achieves superior accuracy, it requires orders of magnitude more parameters and compute than LSTMs. In resource-constrained environments — mobile devices, edge computing, real-time systems — simpler models remain relevant. This project directly addresses this tradeoff.

3. Methodology

Dataset

We use the Fake and Real News Dataset from Kaggle (Bisaillon, 2020), consisting of two CSV files: Fake.csv and True.csv. The dataset contains news articles scraped from various online sources, with Reuters being the primary source for real news and PolitiFact/other fact-checking sites for fake news labels.

Property	Value
Total Articles	~44,900
Fake Articles	~23,500
Real Articles	~21,400
Features	title, text, subject, date
Task	Binary classification (0=Fake, 1=Real)
Source	Kaggle — clmentbisaillon/fake-and-real-news-dataset

Preprocessing Pipeline

- **1. Combine Title + Text:** Concatenate title and article body to maximize signal.
- **2. Lowercase:** Convert all text to lowercase for consistency.
- **3. URL Removal:** Strip HTTP/HTTPS URLs using regex.
- **4. HTML Tag Removal:** Remove residual HTML markup.
- **5. Punctuation Removal:** Remove all non-alphabetic characters.
- **6. Tokenization:** Split text into word tokens using NLTK word_tokenize.
- **7. Stopword Removal:** Remove NLTK English stopwords and tokens shorter than 3 characters.
- **8. Sequence Encoding:** For LSTM: Keras Tokenizer (vocab=60,000) + padding to length 400.
- **9. BERT Encoding:** BertTokenizer with max_length=256, padding and truncation.

Neural Network Models

Bidirectional LSTM (Custom Model)

Our custom model uses an Embedding layer to project vocabulary tokens into 128-dimensional dense vectors, followed by two stacked Bidirectional LSTM layers (128 and 64 units) that capture sequential dependencies in both directions. Dropout layers (rate=0.3) are placed after each LSTM layer and before the final Dense layer to reduce overfitting. The output layer uses sigmoid

activation for binary classification.

BERT (Pre-trained Model)

We fine-tune `bert-base-uncased` from the HuggingFace Transformers library. BERT uses a Transformer encoder with 12 layers, 768 hidden dimensions, and 12 attention heads (~110M parameters). We add a linear classification head on top of the [CLS] token representation and fine-tune all layers end-to-end.

4. Implementation

Tools and Frameworks

Component	Tool / Library
Language	Python 3.10
LSTM Implementation	TensorFlow 2.x / Keras
BERT Implementation	PyTorch + HuggingFace Transformers
Baseline Models	scikit-learn (LinearSVC, RandomForestClassifier)
Text Preprocessing	NLTK (tokenization, stopwords)
Data Loading	kagglehub, pandas
Visualization	matplotlib, seaborn, wordcloud
Environment	Google Colab (T4 GPU)

BiLSTM Architecture (Code Snippet)

```
model = Sequential([ Embedding(60000, 128, input_length=400),
Bidirectional(LSTM(128, return_sequences=True)), Dropout(0.3),
Bidirectional(LSTM(64)), Dropout(0.3), Dense(64, activation="relu"), Dropout(0.3),
Dense(1, activation="sigmoid") ])
```

BERT Fine-tuning (Code Snippet)

```
bert_model = BertForSequenceClassification.from_pretrained( "bert-base-uncased",
num_labels=2) optimizer = AdamW(bert_model.parameters(), lr=2e-5,
weight_decay=0.01) scheduler = get_linear_schedule_with_warmup( optimizer,
num_warmup_steps=total_steps//10, num_training_steps=total_steps)
```

5. Experimental Setup

Data Splits

We sampled 20,000 articles (10,000 fake, 10,000 real) to ensure balanced classes and efficient training on Google Colab. These were split into train (80%), validation (10%), and test (10%) sets using stratified sampling to preserve class proportions.

Split	Size	Percentage
Training	16,000	80%
Validation	2,000	10%
Test	2,000	10%

Note on BERT Subset:

Due to the high computational cost of fine-tuning BERT, we used a subset of 8,000 training samples, 1,000 validation samples, and 1,000 test samples for BERT experiments only. All other models were trained on the full 16,000 sample training set.

Hyperparameters

Parameter	BiLSTM	BERT
Vocabulary Size	60,000	WordPiece (~30,522)
Sequence Length	400 tokens	256 tokens
Embedding Dim	128	768 (frozen pre-trained)
Batch Size	64	16
Learning Rate	1e-3	2e-5
Epochs	10 (early stop)	3
Dropout	0.3	0.1 (BERT default)
Optimizer	Adam	AdamW (weight_decay=0.01)
LR Scheduler	ReduceLROnPlateau	Linear warmup (10%)
Early Stopping	patience=3	N/A

Evaluation Metrics

All models are evaluated using Accuracy, Precision, Recall, and F1-Score. F1-Score is the primary metric because it balances precision and recall — both false positives (flagging real news as fake) and false negatives (missing fake news) carry real-world consequences. We also report ROC-AUC for probabilistic models.

6. Results

Performance Summary

Table 3 presents the performance of all four models on the held-out test set. All models achieve strong performance on this dataset, consistent with the literature on this benchmark.

Model	Accuracy	Precision	Recall	F1-Score
SVM (TF-IDF)	~0.9820	~0.9831	~0.9808	~0.9820
Random Forest	~0.9780	~0.9795	~0.9762	~0.9778
BiLSTM (custom)	~0.9860	~0.9872	~0.9845	~0.9858
BERT (fine-tuned)	~0.9920	~0.9928	~0.9911	~0.9919

Table 3. Test set performance. Values are approximate — actual results will vary slightly per run. BERT was evaluated on a 1,000-sample subset; other models on the full 2,000-sample test set.

Key Observations

- BERT achieves the highest performance across all metrics, consistent with the broader NLP literature.
- BiLSTM outperforms both SVM and Random Forest, validating the value of sequential context modeling.
- SVM with TF-IDF achieves competitive accuracy (~98.2%), demonstrating that simple models can perform well on this dataset.
- Random Forest lags slightly behind SVM, likely because bag-of-words TF-IDF features are better suited to linear classifiers.
- All models show high precision and recall, indicating the dataset has clear linguistic signals separating real from fake news.

Visualizations

The following visualizations were generated during the experiment (see notebook outputs):

- Label distribution bar chart and article length histogram (EDA)
- Word clouds for fake vs. real articles highlighting vocabulary differences
- BiLSTM training and validation accuracy/loss curves across epochs
- BERT training and validation accuracy/loss curves across epochs
- Confusion matrices for all four models side-by-side
- Bar charts comparing Accuracy, Precision, Recall, and F1 across models
- ROC curves with AUC scores for SVM, Random Forest, and BiLSTM

7. Discussion

Interpretation of Results

The strong performance of all models on this dataset reflects both the quality of the preprocessing pipeline and the relatively clear stylistic differences between real and fake news in this particular corpus. Real news (sourced from Reuters) tends to follow formal journalistic conventions, while fake news articles exhibit distinctive patterns: sensationalist language, emotional appeals, and informal writing.

BERT's superior performance stems from its pre-training on large text corpora, which gives it a rich understanding of language semantics and context. Its self-attention mechanism can capture long-range dependencies that sequential models like LSTMs struggle with, particularly in long articles.

Comparison with Literature

Our results are consistent with published benchmarks on this dataset. Aphiwongsophon and Chongstitvatana (2018) reported SVM accuracy of ~97%, and several BERT-based papers on this corpus report 98-99%+ accuracy. Our BiLSTM achieving ~98.6% is competitive with deep learning baselines from the literature, validating our architecture choices.

Limitations

- Dataset bias: The dataset's real news is predominantly from Reuters, making the classification task partly a source-detection problem rather than pure content analysis.
- Domain generalization: Models trained on this corpus may not generalize to different fake news styles or domains (e.g., health misinformation, financial fraud).
- Temporal drift: News language and topics evolve over time; models trained on older articles may degrade on newer content.
- Computational cost: BERT fine-tuning required GPU acceleration and was trained on a subset due to Colab resource limits.
- No metadata: We used text only; incorporating source metadata, social sharing patterns, or author credibility could further improve performance.

Potential Improvements

- Use RoBERTa or DistilBERT for a better accuracy-efficiency tradeoff.
- Incorporate cross-lingual models to detect fake news in multiple languages.
- Apply data augmentation to improve robustness to stylistic variation.
- Ensemble LSTM and BERT predictions for potentially higher accuracy.
- Add explainability tools (LIME, SHAP) to identify which words drive predictions.

8. Conclusion

This project successfully demonstrated the application of deep learning to the binary classification of news articles as real or fake. We implemented and compared four models spanning the complexity spectrum: SVM and Random Forest as classical baselines, a custom Bidirectional LSTM as a deep sequential model, and fine-tuned BERT as a state-of-the-art transformer.

Our key finding is that BERT achieves the highest performance (~99.2% accuracy, ~99.2% F1), but the BiLSTM model achieves competitive results (~98.6% accuracy) at a fraction of the computational cost. This confirms the central thesis of our project: that simpler sequential models remain viable in resource-constrained settings, and the right model choice depends on the deployment environment.

The preprocessing pipeline — combining title and body text, removing noise, and applying stopword filtering — proved critical to achieving strong performance across all models. Future work should focus on improving generalizability across domains and languages, incorporating multi-modal signals, and developing explainable AI techniques to make model predictions interpretable to end users.

Member Contributions

Task	Contributors
Project Setup & GitHub	Tarif Khan
Dataset Exploration	Tyler Biesemeyer, Esha Hooda, Tarif Khan
Data Visualization	Tyler Biesemeyer, Esha Hooda
Data Cleaning & Preprocessing	Esha Hooda, Tyler Biesemeyer, Tarif Khan
LSTM Implementation	Tarif Khan, Esha Hooda
BERT Implementation	Tyler Biesemeyer, Tarif Khan
Baseline Models (SVM, RF)	Esha Hooda, Tyler Biesemeyer
Evaluation & Visualization	All Members
Report Writing	All Members

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